Fisheries & aquaculture

Diseases, Management, & Animal Welfare

By:

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Aim of this lecture

The student will be aware of on farm fish behavior deviations that are indicative of diseases.

By the End of this lecture the student will be aware of common fish diseases, causative agents and treatments.

The student will be aware of best management pratices for prevention and Mitigation of fish diseases.

Outline

- Myself
- Key terms in fish diseases
- Introduction to fish diseases (host, environment, pathogen interactions)
- Diagnosis and Fish behaviour to detect disease in aquaculture
- Common fish diseases(Bacteria, parasites, viruses)
- Prevention and Treatment
- Recent advances in global fish diseases, prevention, diagnostic and treatment.

- BSc. Fisheries & Aquaculture~2017
- M.Sc. Aquaculture~2022
- M.Sc. Envi. Sci. Limnology and wetland Management~ 2024
- PhD researcher ~ presently



Fellowship, Network, Membership

- Mandela Washington fellows for young African leaders

- **Austria Federal Agency for Water Management~Present**
- **WorldFish CGIAR centre**: 2020 -2022.
  Technical officer fish breeding and genetics.
- **Savanna Streams Limited:** 2017 to 2019.
  Aquaculturist

Experiences from:

- Zambia, Malawi, USA,
- Austria, Netherlands, France
- Kenya

Key terms in fish diseases

- **Abdomen:** The part of the crustacean body that lies behind the genital segments; the last
- **Epizootic:** An outbreak of a disease or an unusually large increase in prevalence and/orintensity of a parasite in an animal population.
- **Exophthalmus**: Protrusion of the eye.
- **Formalin**: A 37% solution of formaldehyde.
- **Hypoxia**: The reduction of oxygen supply to tissues below physiological levels despite adequate perfusion of the tissue by blood

Key terms in fish diseases

- **Dropsy:** is a condition in fish caused by the buildup of fluid inside the body cavity or tissues. As a symptom rather than a disease in its own right, it can indicate a number of underlying diseases, including bacterial infections, parasitic infections, or liver dysfunction
- **Acclimation:** the process through which fish become fully adapted to new environmental circumstances; such as being placed into water of different quality, temperature, or different holding situations
- **Ectoparasite:** a parasite that lives on the external surface of the host
- **Hemorrhage:** an escape of blood from the vessels, either through intact blood vessel walls or through ruptured vessels

Key terms in fish diseases

- **Lesions:** are areas of abnormal tissue change due to disease or injury
- **peritoneal cavity:** in fish, as in other vertebrates, is a fluid-filled space within the abdomen that houses and protects internal organs. It is lined by a membrane called the peritoneum
- **Quarantinee:** the practice of isolating newly acquired or potentially diseased aquatic organisms to prevent the spread of diseases to healthy populations.
- **Biosecurity:** in fisheries and aquaculture refers to a set of preventive measures designed to reduce the risk of introducing and spreading infectious diseases, parasites, and invasive species within and between aquatic environments
- **Moribund:** obviously progressing towards death, nearly dead.
- **Lordosis:** dorso-ventral curvature of the spine.
- **Facultative fish pathogens**: occurring naturally as non-pathogens in the environment but capable of causing disease outbreaks under conditions of stress

Key terms in fish diseases

- **Active Immunization:** This involves stimulating the fish's immune system to produce a specific immune response against a pathogen. This method helps fish develop long-term immunity and memory against specific diseases.

- **Passive Immunization:** This involves the direct transfer of antibodies or immune cells to fish, providing immediate but temporary protection against diseases. Passive immunization does not stimulate the fish's immune system to produce its own antibodies**. It is useful for providing short-term** immunity.

- **Immunostimulants:** substances that enhance the immune response of fish, making them more resistant to infections. Unlike vaccines, immunostimulants do not target specific pathogens but boost the overall immune system.

- **Probiotics:** live microorganisms that, when administered in adequate amounts, confer a health benefit on the host. Used to improve **the gut health** of fish, **enhance nutrient absorption**, and o**utcompete** pathogenic bacteria.

Introduction to fish diseases

What is disease?

- Disease, any *harmful deviation* from the normal structural or functional state of an organism, generally associated with certain signs and symptoms and differing in nature from physical injury.

- How do you know of a disease:

You need to know the normal nature of an organism

Healthy Fish Behaviour

Behavioural indicators of welfare in farmed fish
~ Catalina et al..2012

Qualitative assessment

1. Good food anticipatory behaviour
2. Good feed intake
3. Smooth swimming behaviour
4. Proper Ventilation

1. Poor ~ less 10% Food eaten

2. Fair ~20 % Food eaten

3. Good ~50 % Food eaten

4. Very Good ~ 80 % Food eaten

5. Excellent ~ 100 % Food eaten

Things to consider: Time given to the fish to eat the feed, ration size, and duration of last feeding

Fish behaviour to detect disease

- Changes in behaviour and posture drifting, whirling
- Rubbing/flashing, jumping
- Fin clamping
- poor body condition proportion of head to body
- Respiratory rate hovering, piping (gulping air),
- Externally visible lesions/changes
- Body shape
- Discolouration
- Eyes
- Dropsy, protrusion of scales, darkening, pop eyes



What will cause disease in Fish

1. **Bacterial diseases**: They are the diseases that caused by bacterial agent such as *Furunculosis, Bacterial kidney Disease, Columnaris*, etc.
2. **Viral Diseases:** They are the diseases that caused by viral agent such as Infectious Pancreatic Necrosis, Viral *Haemorrhagic Septicaemia*, Tilapia lake virus
3. **Mycotic Diseases**: They are the diseases that caused by fungal agent such as *Saprolegniosis, Branchiomycosis,* etc.
4. **Parasitic Diseases:** They are the diseases that are caused by parasitic agent.
5. **Protozoa:** such as *Ichthyobodoosis*. *Hexamitosis,* Coccidiosis, Whirling diseases

Other Diseases

Nutritional Diseases

- Essential amino acids :lysine, methionine, tryptophan, arginine, histidine, leucine, isoleucine.

- Essential fatty acids :18:2 omega-6, 18:3 omega-3; 20:4 omega-6, 20:5 omega-3, 22:6 omega-3 depending upon species.

- Vitamins :B1, B2, B6, B12, pantothenic acid, niacin, biotin, folicacid, in ositol, choline, D3, A, K3, E, and C



Figure 2. This fish has a slight degree of scoliosis, a sign of possible malnutrition (e.g., hypovitaminosis C). © *Jörg Mayer*

Other Diseases

- Neoplastic diseases



(a)

Fish Cancer

The uncontrolled growth of cells within a living organism, which disrupts the normal tissue architecture. Tumor refers to the resulting gross or histologic mass of neoplastic cells.

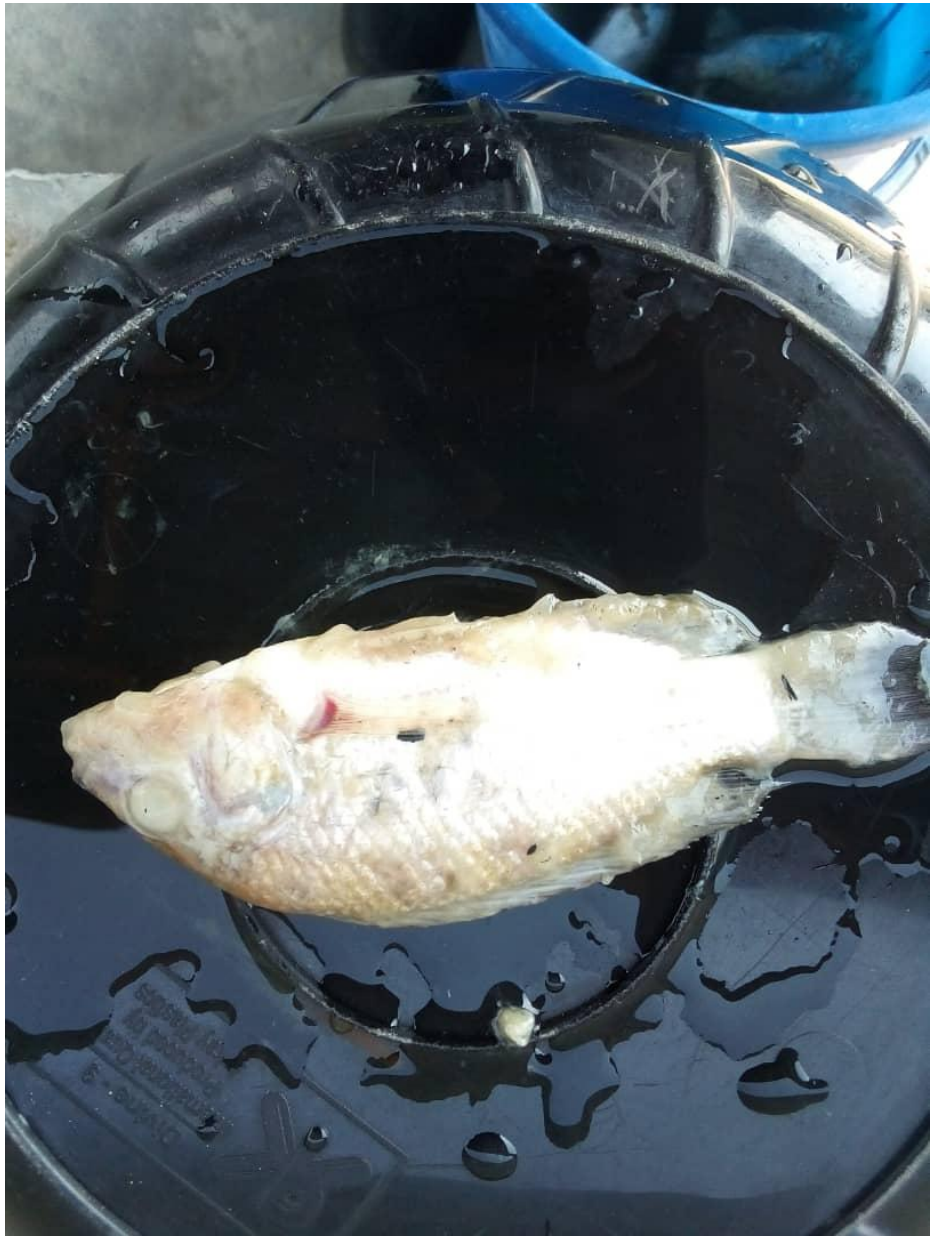
Ferraro, E.; Harrison, S.H.; Duke, E.; Troan, B.; Boddy, A.; Abegglen, L.M.; Harrison, T.M. Retrospective Study of the Prevalence, Histopathology, Therapy, and Survival Time of Neoplastic Disease in Fish. Animals 2024, 14, 464. https://doi.org/10.3390/ani14030464

Other Diseases

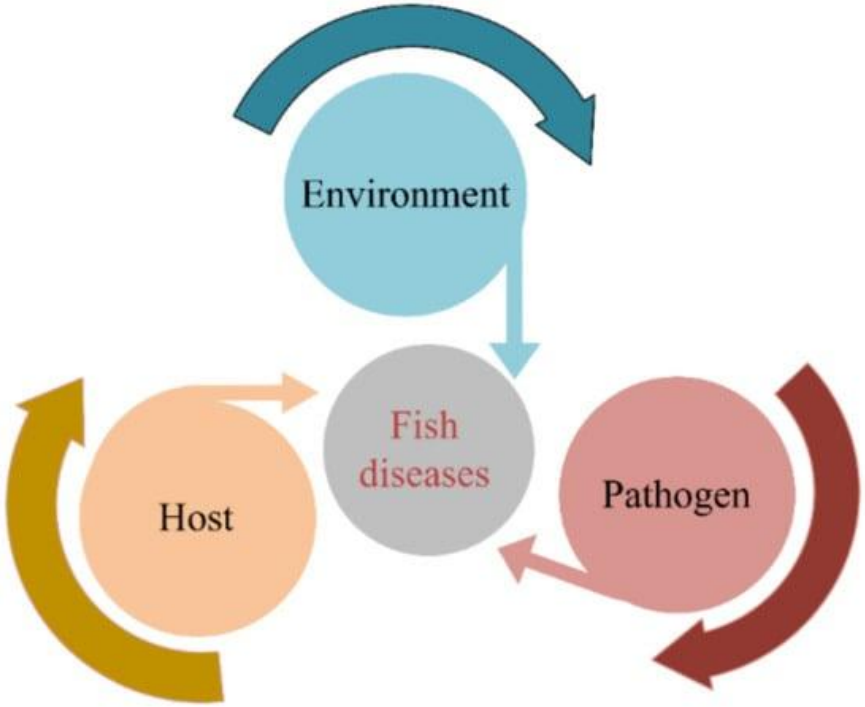
- Fungal infections

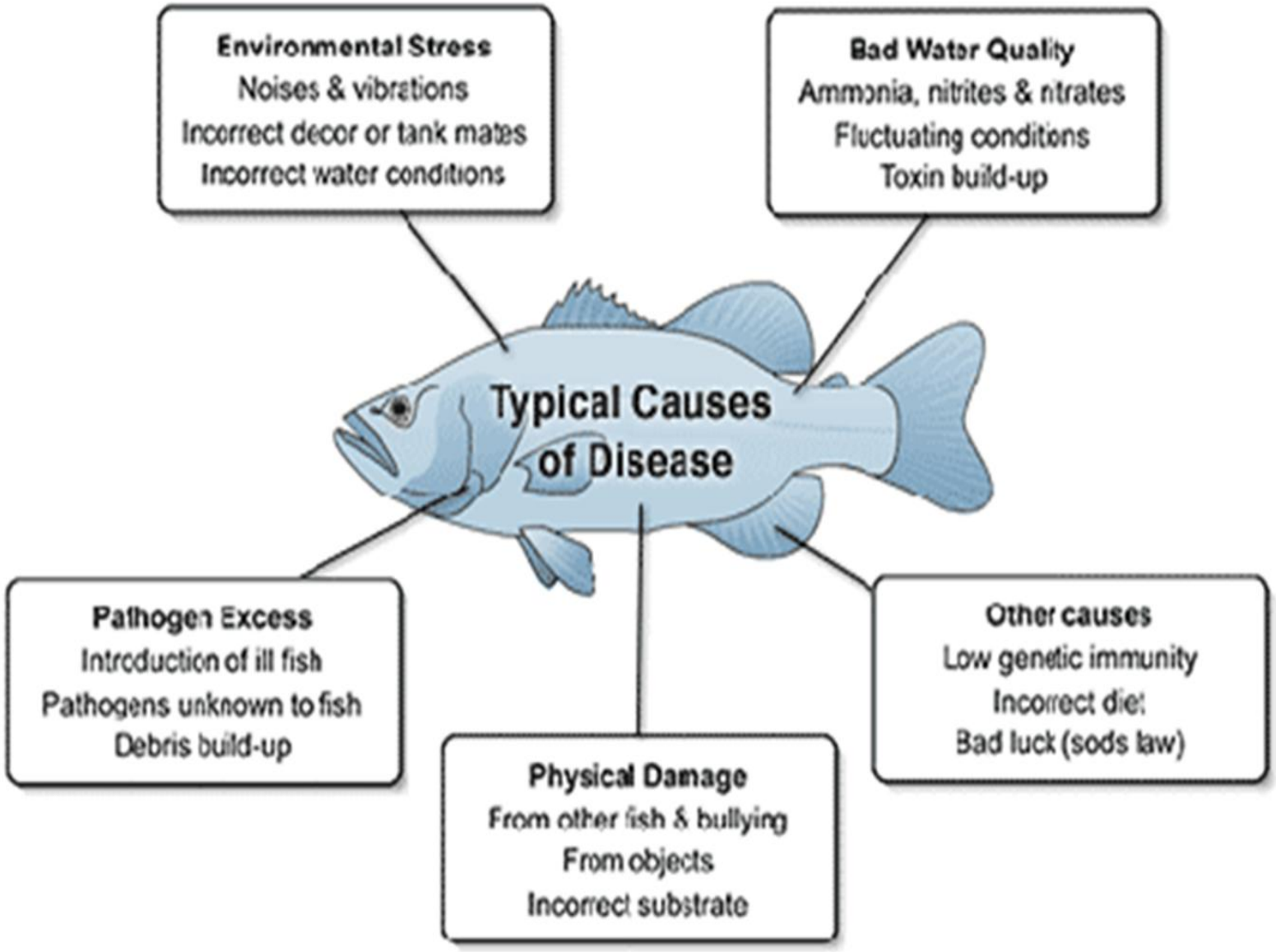
- Common in the cold season





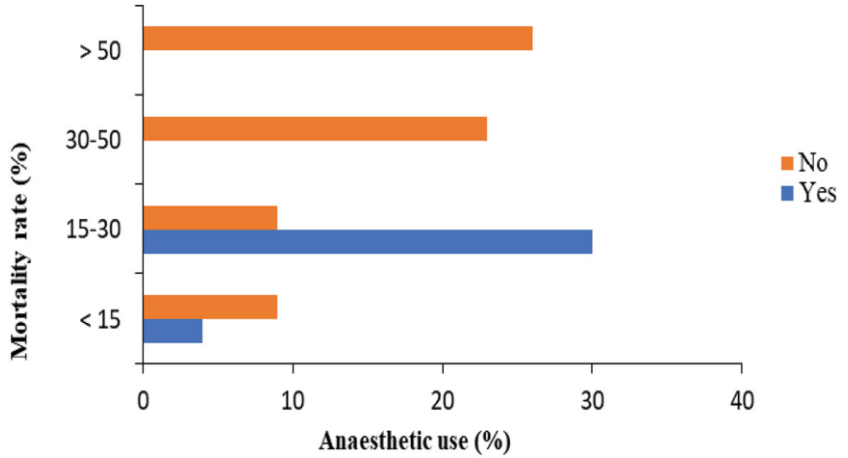
What causes fish disease





What will cause disease in Fish

Stress



- Water quality deuteriation
- **Handling**

Use of anaesthetics is important to prevent stress-related diseases due to handling.

The highest mortality rates experienced by the hatcheries that applied anaesthetics were in the range of 15–30%, while those that did not use anaesthetics experienced higher mortality rates (>50%).

Mphande, J., Hasimuna, O. J., Kikamba, E., Maulu, S., Nawanzi, K., Phiri, D., ... Siavwapa, S. (2023). Application of anaesthetics in fish hatcheries to promote broodstock and fish seed welfare in Zambia. Cogent Food & Agriculture, 9(1). https://doi.org/10.1080/23311932.2023.2211845

Common diseases

Especially in subsaharan countries

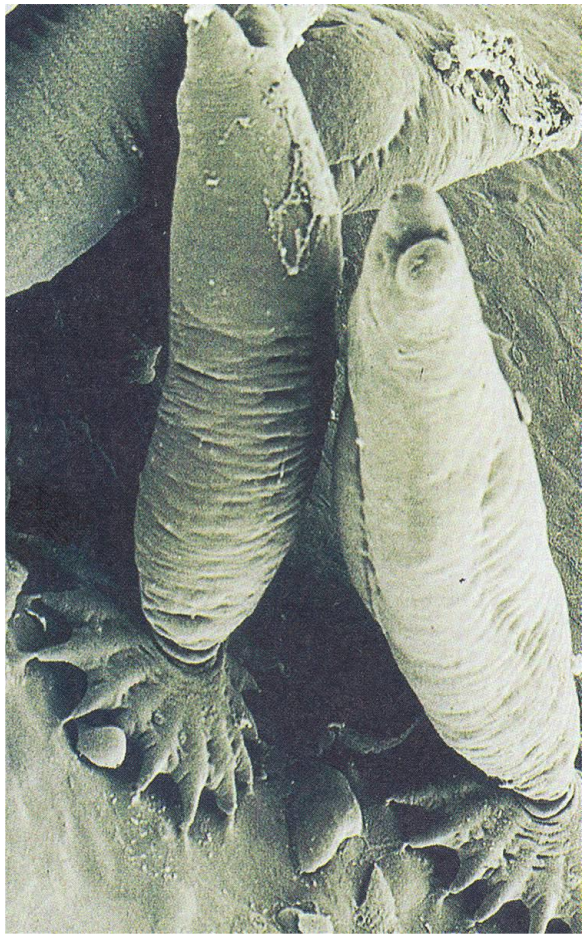


Common fish diseases parasites

Monogeneans

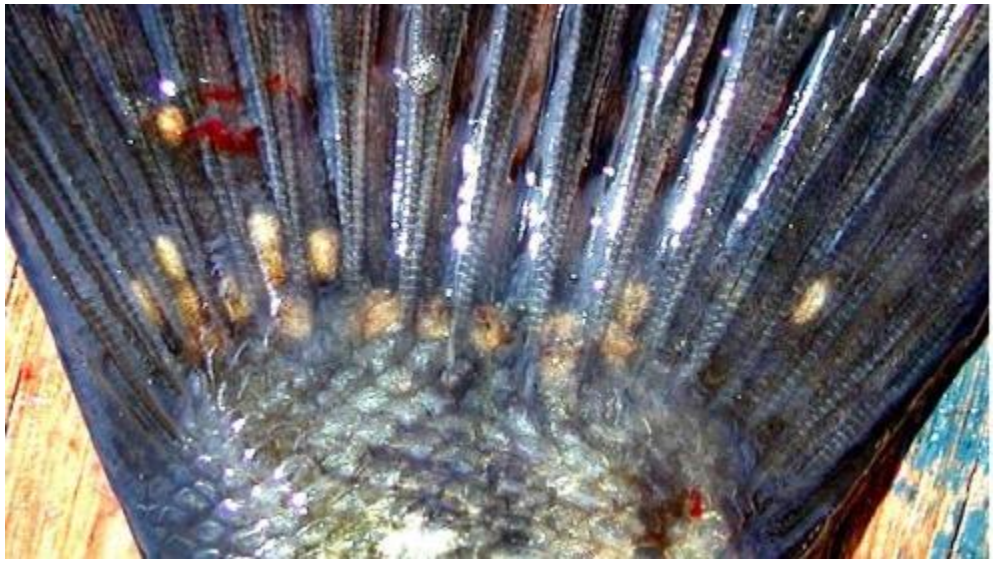
- are parasitic flatworms that are commonly found on fishes and lower aquatic invertebrates

- Monogeneans have a direct life cycle, which means they go directly from host to host (fish to fish)





Parasites Yellow grub *(Clinostomum complanatum)*



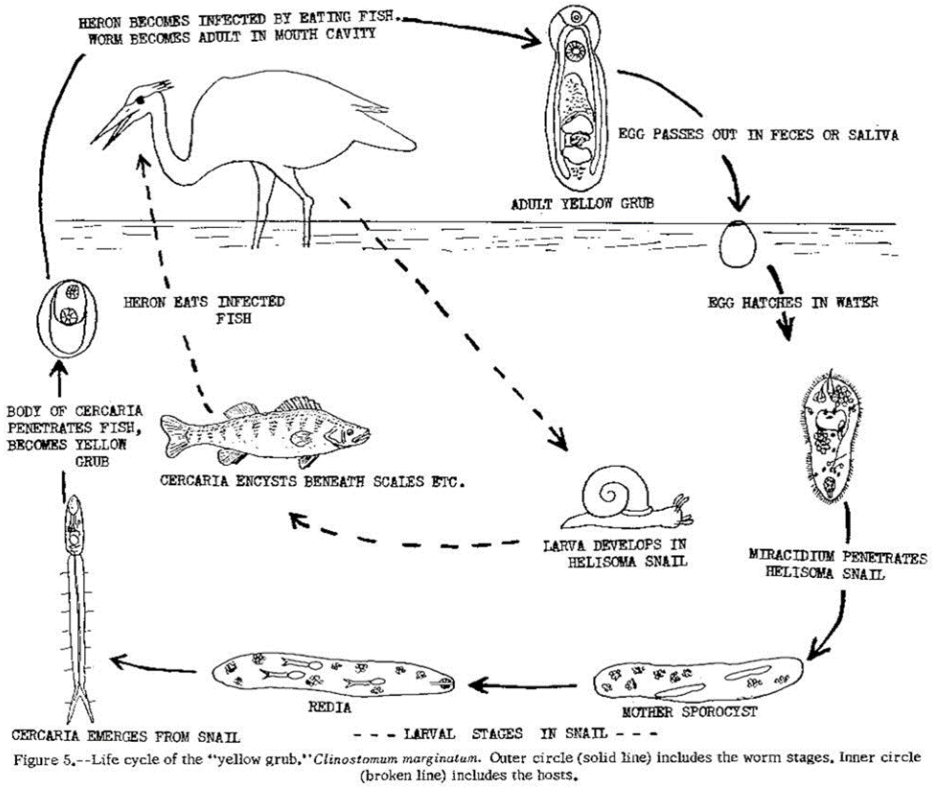
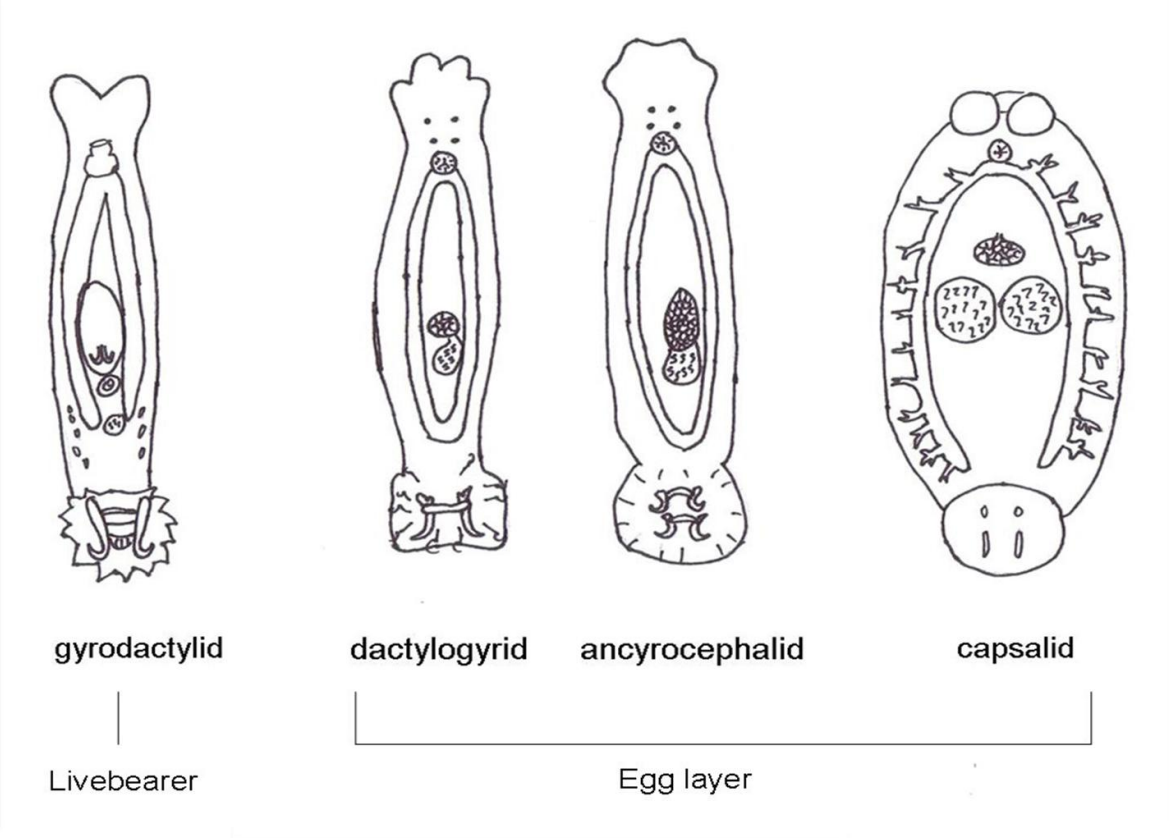


Figure 5.--Life cycle of the "yellow grub," *Clinostomum marginatum*. Outer circle (solid line) includes the worm stages. Inner circle (broken line) includes the hosts.

- large (3 to 8 mm).
- Yellow grubs are embedded intermuscularly (in the muscle) or subcutaneously (under the skin) in fish.

Monogeans

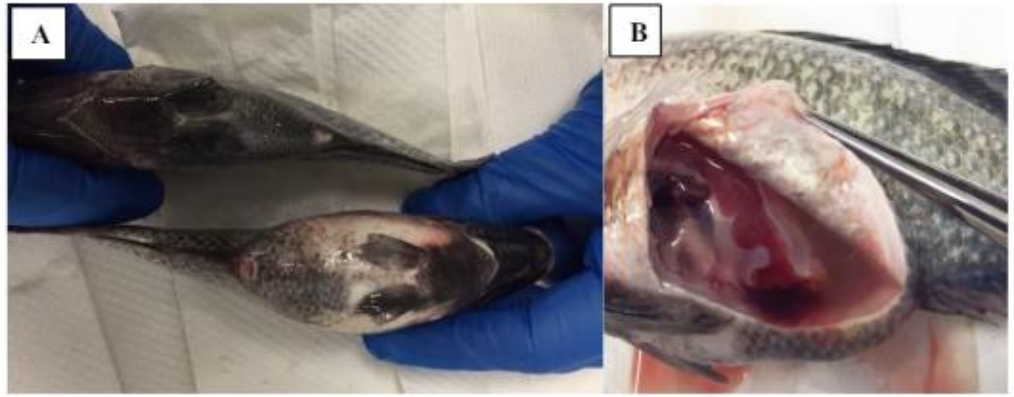




- Fish gasps for air at the water's surface
- Gills open and close rapidly
- Gills are covered in mucus
- Part of gills may be eaten away

Bacteria

- Streptococcosis is primarily caused by the **Gram-positive** bacteria *Streptococcus iniae*



'igure 5. Nile tilapia infected with *Streptococcus agalactiae* Ib showing abdominal distention (A) and bloody fluid in the peritoneal cavity (B).



'igure 6. Hemorrhage from the vent of Atlantic menhaden with a streptococcosis. Photographs taken from Shotts and Plumb (1994) with permission.

1.3.3 Streptococcosis - 4

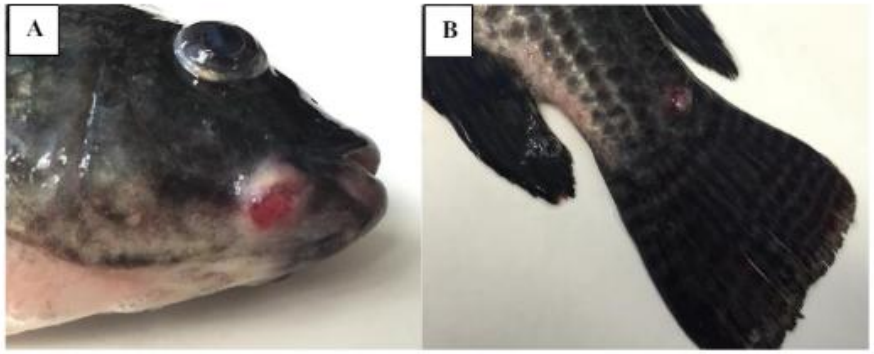


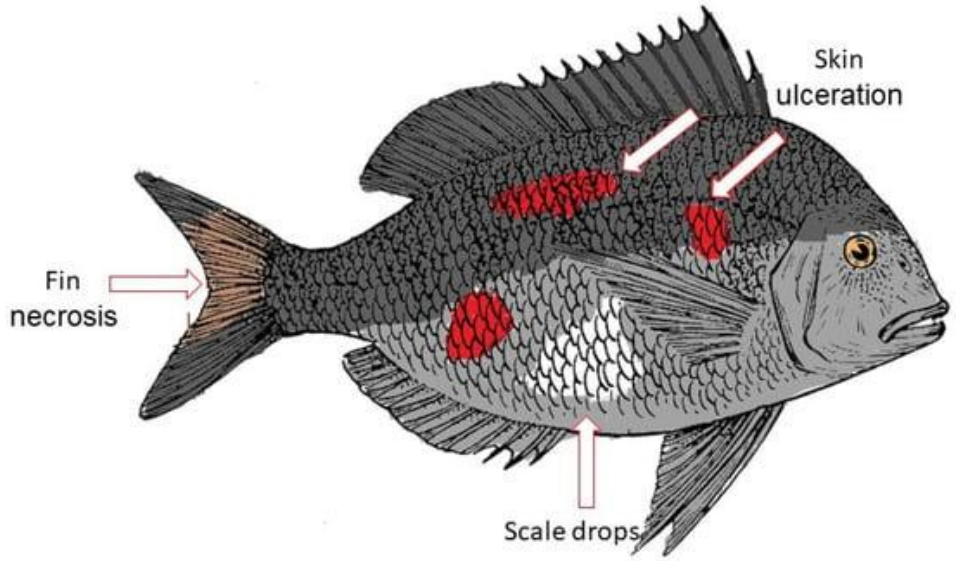
Figure 4. Nile tilapia surviving a *Streptococcus iniae* challenge (by intraperitoneal injection) showing jaw pustules (A) and pustules on the caudal peduncle (B). Tilapia challenged by intramuscular injection with *S. agalactiae* Ib exhibit similar lesions.

Craig A. Shoemaker, Benjamin R. LaFrentz, Julio C. García, Dehai Xu and Benjamin H. Beck United States Department of Agriculture, Agricultural Research Service Aquatic Animal Health Research Unit

Bacteria



Motile aeromonad disease



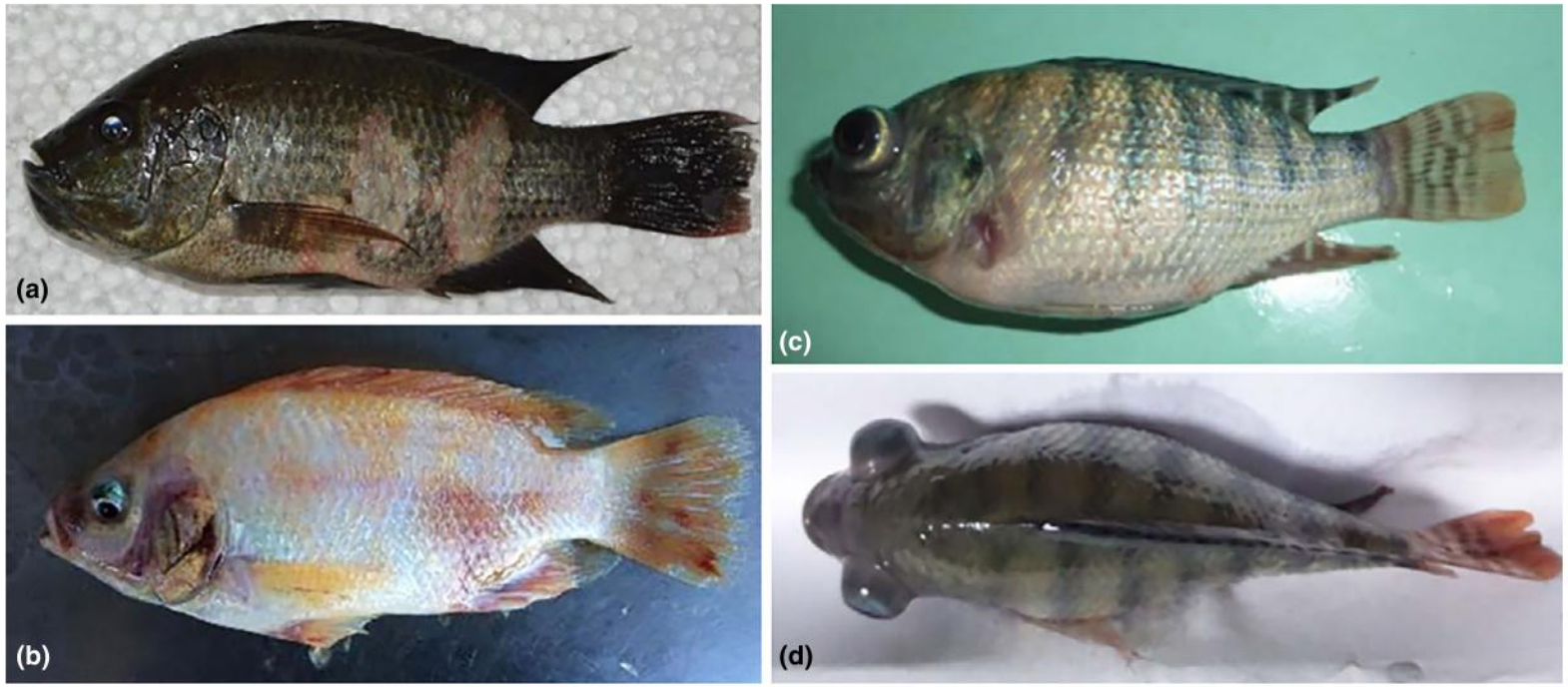
Vibriosis

Other diseases

- Furunculosis
- Bacteria kidney disease
- Bacteria gill disease
- Mycobacteriosis of fishes (TB)

Viral disease

- Tilapia Lake Virus is an emerging viral disease associated with high morbidity and mortality in tilapia.
- To date, TiLV has been reported in 15 countries.
- Multiple infections of TiLV, bacteria and parasites are commonly found in Summer mortality, or One month mortality syndrome in tilapia.



Tilapia lake virus: a threat to the global tilapia industry? 2019
Mona Dverdal Jansen , Ha Thanh Dong2 and Chadag Vishnumurthy Mohan

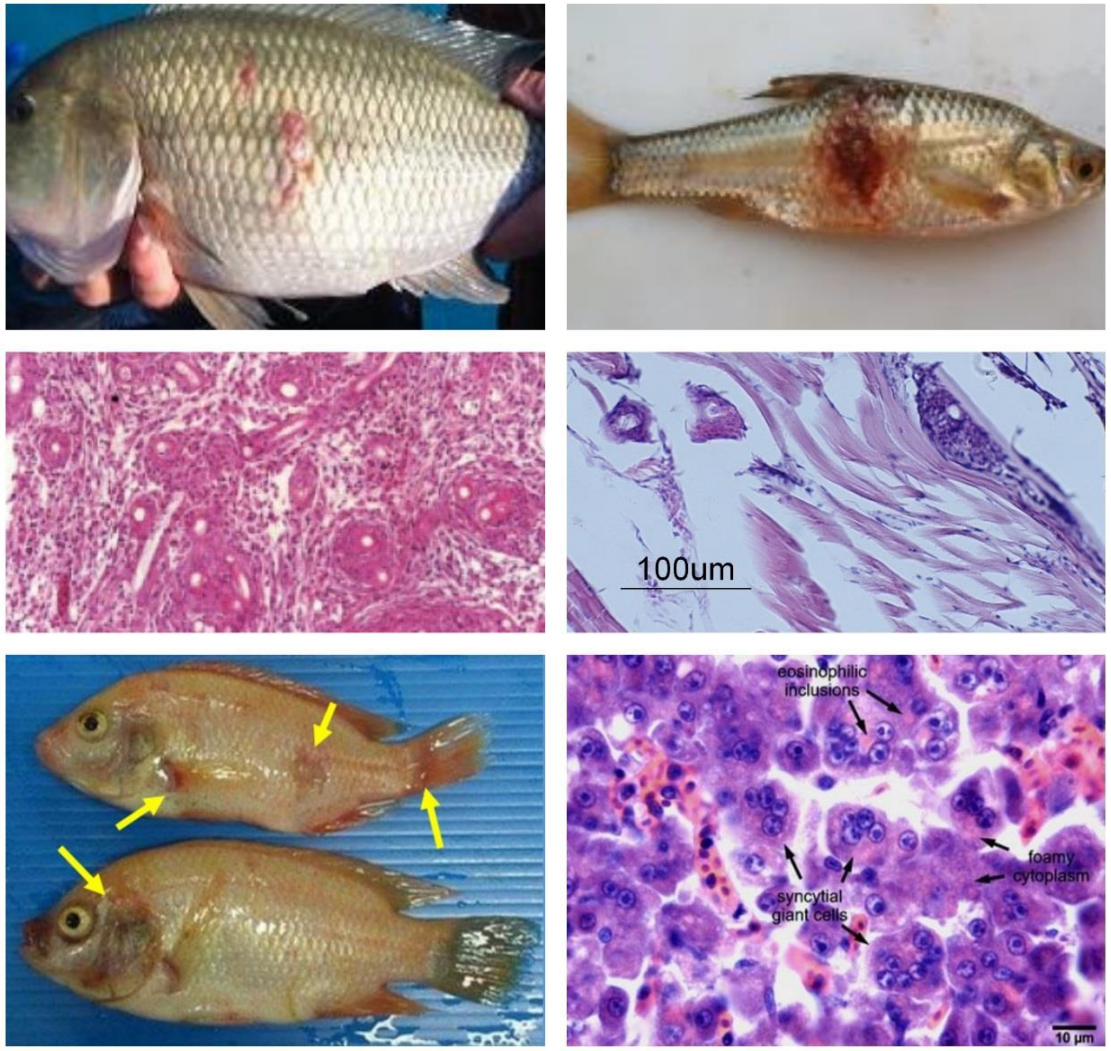
Water mold disease

Epizootic ulcerative syndrome (EUS)

(EUS) is a severely infectious disease of fish that is known to affect more than 100 types of wild, farmed freshwater and estuarine fish, and has historically caused serious losses in the aquaculture industry.

Causative Agent: The primary pathogen associated with EUS is a water mold called Aphanomyces invadans (*also known as Aphanomyces piscicida*).

This pathogen invades fish tissue, leading to extensive necrosis



Technical Assistance to the Zambia Aquaculture Enterprise Development Project (ZAEDP): Output 4: Improving Aquatic Animal Health

Treatment of common fish diseases in aquaculture

Parasitic diseases treatment

- Potassium permanganate dip (Banned)
- Formalin dip
- Salt

Selected therapy for ectoparasites on fish

compound	Dose level	Time and method	Result
Formaline	250mg/L	hour bath 1	No oftener than every 3rd day
Potassium permanganate	2.5mg/L	indefinite	Used as an ulternate to formalin in pond
Sodium chloride	2.5%	indefinite	For fish acceptable to high salinity
Hydrogen peroxide	0.05%	minutes dip 20	Used under strect precautions
Acitic acid	5%	minute dip 1	Useful for aquarium fishes

Treatment of common fish diseases in aquaculture

Bacteria disease treatment

- Potassium permanganate ($KMnO_4$) can be used at 5 mg/L for one hour. Treatment for three days is recommended for “flow through” water system.

- Antibiotics (has led to the widespread emergence of antibiotic resistance)

- Vaccines

Viruses disease treatment

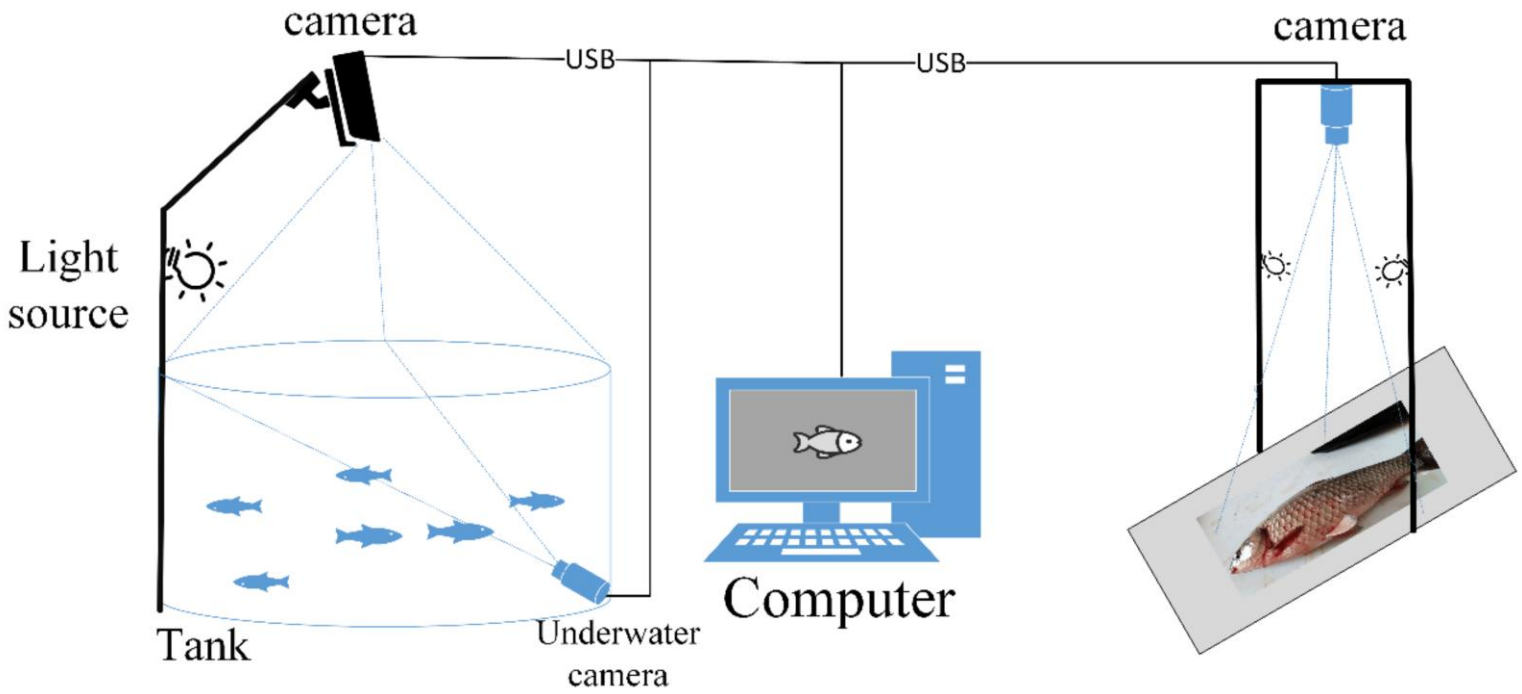
- **Vacinees**
  - Injection vaccination
  - Immersion vaccination

Oral vaccination

The oral vaccination method is considered as one of the methods for vaccinating fish in which the vaccine is first incorporated into the feed and then fed to fish.

Advances in global fish diseases, prevention, diagnostic and treatment

- **Diagonsitic**



Pollution Detection

Pathogen Detection Based on Spectral Images

Diagnosis of Internal Tissues Based on Microscopic Images

Advances in global fish diseases, prevention, diagnostic and treatment

- **Molecular Diagonsitic**

molecular approaches





Polymerase chain reaction tests



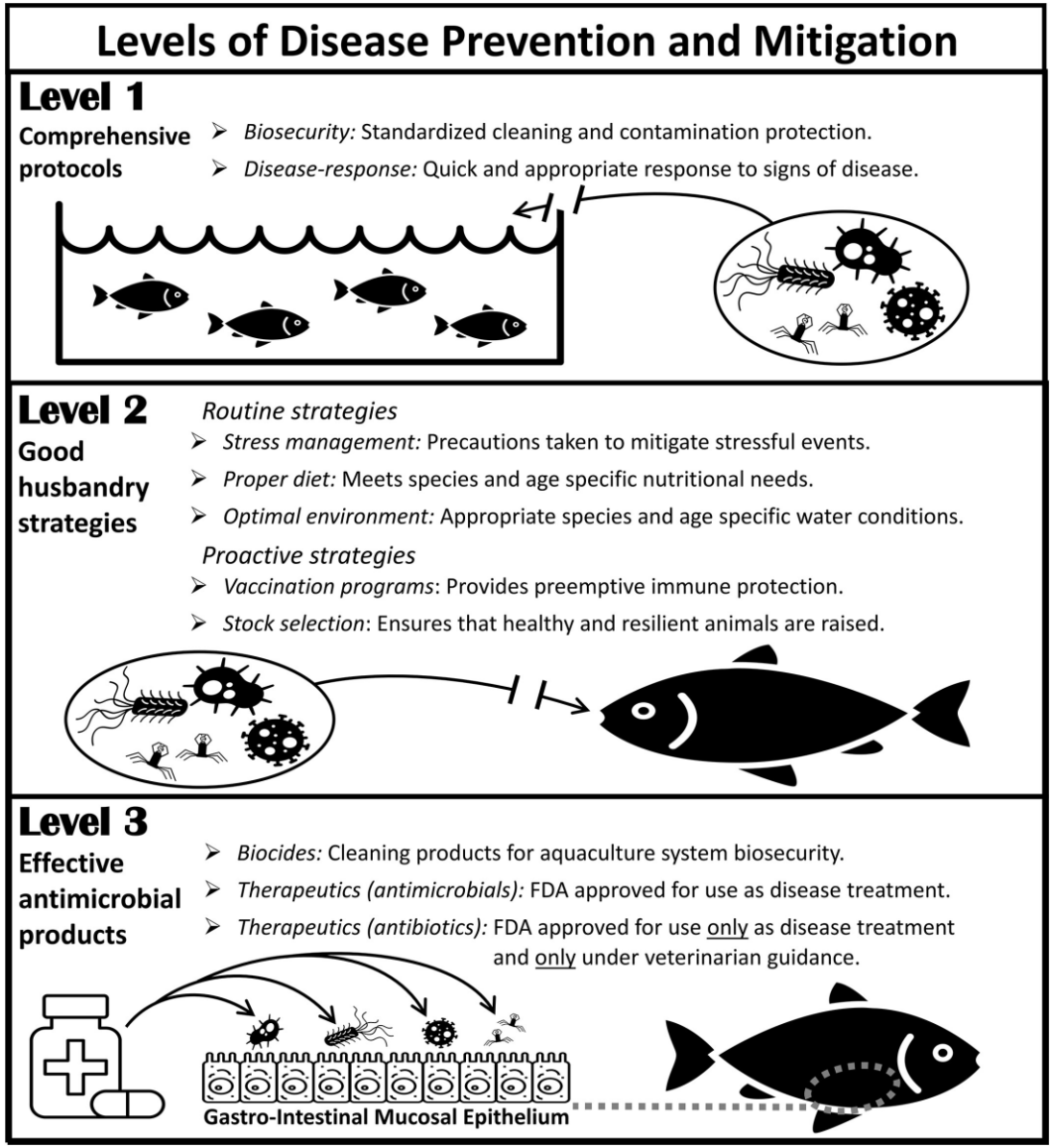
CRISPR, lateral flow immunoassays (LFIAs) and **next-generation sequencing (NGS)** are some of the molecular diagnostic technologies currently available.

Advances in global fish diseases, diagnostic and prevention

- Prevention
- Active and Passive Immunization
- Immunostimulants
- Selective breeding
- Probiotics

Aquaculture & Fisheries management for disease prevention

- Feed management
- Water quality management
- Biosecurty
- Quarantine



Aquaculture & Fisheries management for disease prevention ~**Zambia**

Biosecurity Measures
✓ Fish inspection
✓ Record of visitors to the farm
✓ Personal protective equipment (PPE)
✓ Footbaths
✓ Wheel dip
✓ Wildlife control
✓ Staff awareness
✓ Good quality feed
✓ Water source and routine water management
✓ Farm practices and records
✓ Emergency response
✓ Handling of mortalities
✓ Disposal of mortalities

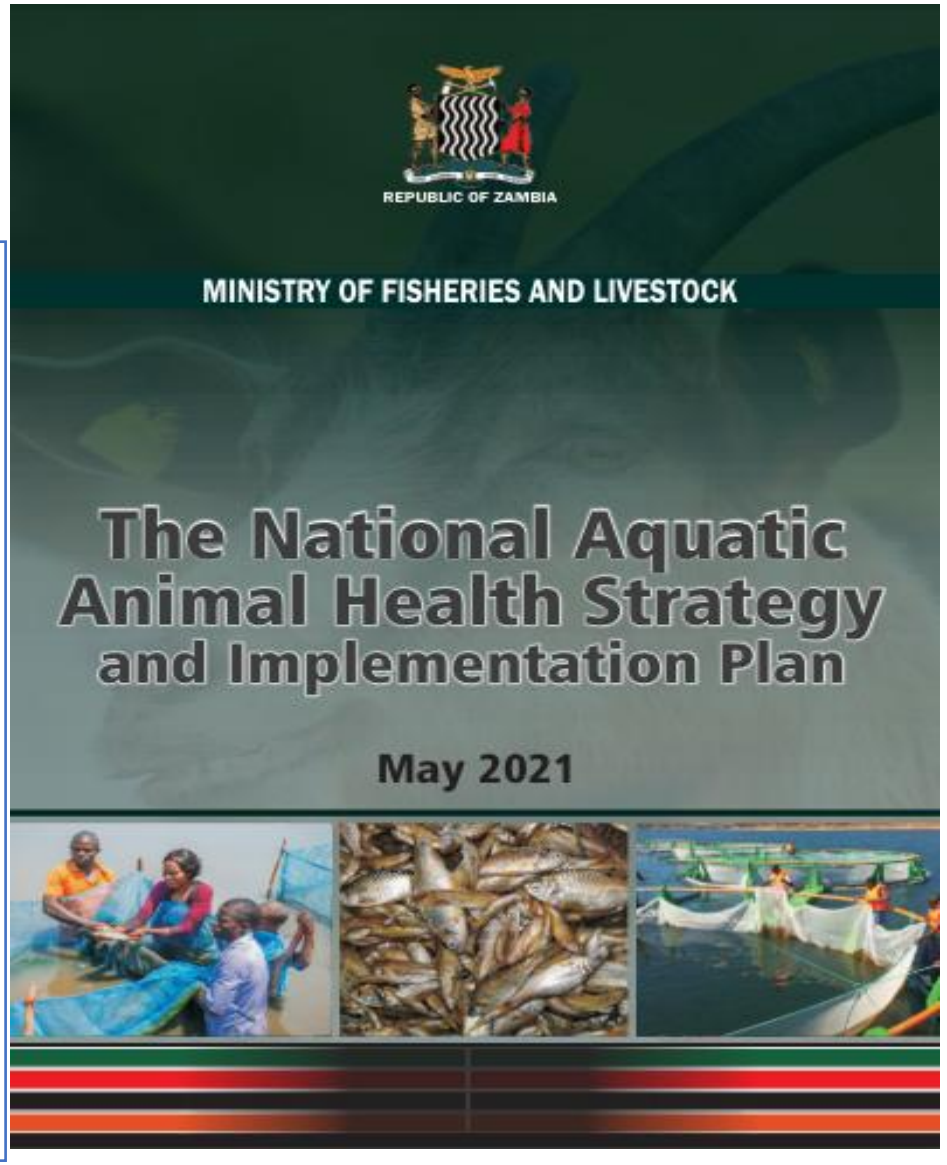


Basiita K. R, Malambo T, Hampuwo B, Chungu P and Songe M. 2022. Biosecurity practices for tilapia hatcheries: A case of Zambia. Penang, Malaysia: WorldFish. Fact Sheet: 2022-08.

Aquaculture & Fisheries management for disease prevention~**Zambia**

Strategic Objectives:

- Enhance Aquatic Animal Health Management.
- Enhance AAH Research and Development.
- Enhance collaboration with stakeholders.
- Improve the AAH institutional capacity.
- Operationalise AAH laboratories and support services.



- **End any questions**